

Firman Maulana Nur Alif

Embedded Systems, IoT & Full-Stack Developer

Indramayu, West Java, Indonesia • firman@example.com • github.com/firman • Roblox Developer

PROFESSIONAL SUMMARY

Technology enthusiast specializing in Embedded Systems, IoT, Industrial Automation, and Interactive Software Development. Proven capability in designing ESP32/ESP8266 microcontroller hardware, real-time MQTT communication, OTA firmware solutions, Node.js backend architectures, and advanced Lua-based Roblox experiences. Passionate about bridging hardware automation with robust web services.

TECHNICAL SKILLS

Programming:	JavaScript, Node.js, Lua, C/C++ (Arduino, ESP-IDF)
Embedded & IoT:	ESP32, ESP8266, Arduino, MQTT, OTA Updates, Wokwi
Web & Backend:	HTML5, CSS3, JavaScript, Express.js, REST APIs, WebSockets
Industrial Automation:	PLC, HMI, Modbus TCP/IP, Sensors Integration
DevOps & Networking:	Git, GitHub, Cloudflare Tunnel, VPS, DNS Management
Tools & Platforms:	Roblox Studio, VS Code, Postman

FEATURED PROJECTS

SIPETARUNG: IoT Bird Repellent & Agricultural Monitoring System

ESP8266 / Audio Automation

- Designed an automated agricultural monitoring and bird deterrent prototype utilizing ESP8266 microcontrollers and automated audio frequencies.
- Integrated environmental sensor telemetry to trigger intelligent deterrent routines and remote dashboard visualization.

ESP32 OTA Firmware Update Platform

C++ / Web Interface

- Created a dedicated firmware update platform allowing seamless, remote Over-The-Air (OTA) updates for deployed ESP32 nodes via a responsive web dashboard.
- Streamlined device maintenance cycles and eliminated physical serial re-flashing overhead.

TikTok Live to Roblox Real-Time Integration Bridge

Node.js / Lua / REST API

- Engineered a robust Node.js backend bridge capturing live TikTok event streams and translating them into dynamic Roblox gameplay interactions in real time.
- Optimized socket communication latency to maintain high throughput during high-traffic streaming sessions.

ASMR Keyboard System for Roblox

Lua / Roblox Studio

- Developed a high-performance client-side keyboard interaction and audio feedback system featuring multiplayer synchronization and customizable UI themes.
- Published and maintained as a modular asset for independent Roblox developers.

ESP32 Environmental Monitoring Dashboard

Node.js / MQTT / Charting

- Built an end-to-end telemetry dashboard displaying real-time temperature, humidity, water quality parameters, and industrial sensor metrics.

EXPERIENCE & LEADERSHIP

Roblox Developer & Community Asset Creator

Independent

- Published reusable development assets and interactive systems, serving multiple active developer projects within the ecosystem.

Robotics Extracurricular Instructor

Educational Mentorship

- Transitioned from active robotics team member to technical instructor, guiding junior students through circuit design, microcontroller programming, and automation fundamentals.

EDUCATION & LANGUAGES

Politeknik Negeri Indramayu (Polindra)
D4 Teknologi Rekayasa Instrumentasi dan Kontrol (TRIK)
SMK Negeri 1 Sindang Indramayu
High School / Vocational Education

Languages
• Indonesian: Native
• English: Intermediate